

NAME: \_\_\_\_\_ MATRIC NUMBER: \_\_\_\_\_

**SCHOOL OF MATERIALS SCIENCE & ENGINEERING**

**CONTINUOUS ASSESSMENT 1 SEMESTER 2 2022-2023**

**MS1016 – THERMODYNAMICS OF MATERIAL**

21 February 2023

Time Allowed: 1 hour

**INSTRUCTIONS**

1. This paper contains **THREE (3)** short questions and **TEN (10)** MCQ questions.
  2. Answer **ALL** questions.
  3. This is an open book test.
  4. All questions have **ONLY ONE** correct answer for the MCQ questions.
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**PART I – SHORT QUESTIONS**

1. Explain what you understand by an isothermal process and what are the essential conditions for this process. Describe an example of this process in a living organism.  
(20 marks)

An isothermal process is one in which the temperature of the system remains constant, i.e.  $\Delta T = 0$  during the process. This is achieved by absorbing heat or releasing heat during the process.

The essential conditions are (1) the system must be thermally conducting to the surrounding and (2) the process must be carried out very slowly (reversible) such that there is sufficient time for the heat to be transferred while maintaining a constant temperature.

Example of a living organism is the hot blooded mammals where the body temperature is kept constant by perspiration or by respiratory action.

2. Explain, briefly the concept of the second law of thermodynamics. Also explain the relationship between entropy, work and efficiency.

(20 marks)TT

The second law of thermodynamics states that for a spontaneous event to take place, it must be accompanied by an increase in the entropy.

A system can lower its entropy spontaneously by interacting with the surrounding or with another system but it must be accompanied by an increase in the entropy of the combined systems.

Therefore  $dS_{\text{system}} + dS_{\text{surrounding}} > 0$

In the special case where the interaction is perfectly reversible,

$$dS_{\text{system}} + dS_{\text{surrounding}} = 0$$

For an isolated system, entropy change in a process is given as

$$dS_{\text{system}} = \delta Q/T$$

Therefore for a reversible process, there is no net increase in the combined entropy. For a spontaneous process which is irreversible,

$$dS_{\text{irreversible}} > dS_{\text{reversible}}$$

Increase in the entropy means more heat is required to perform the same amount of work. Therefore, irreversible work results in a loss in the efficiency.

3. Letting  $U = U(T, V)$ , show that for an isolated system,

$$\delta Q = \left( \frac{\partial U}{\partial T} \right)_V dT + \left[ P + \left( \frac{\partial U}{\partial V} \right)_T \right] dV$$

where  $Q$  is the heat,  $T$ , the temperature,  $U$ , the internal energy,  $P$ , the pressure and  $V$ , the volume.

(20 marks)

Let  $U = U(T, V)$

$$\therefore dU = \left( \frac{\partial U}{\partial T} \right)_V dT + \left( \frac{\partial U}{\partial V} \right)_T dV$$

But from 1<sup>st</sup> Law of Thermodynamics,

$$dU = \delta Q - PdV$$

Substituting,

$$\delta Q - PdV = \left( \frac{\partial U}{\partial T} \right)_V dT + \left( \frac{\partial U}{\partial V} \right)_T dV$$

$$\delta Q = \left( \frac{\partial U}{\partial T} \right)_V dT + \left( \frac{\partial U}{\partial V} \right)_T dV + PdV$$

$$\delta Q = \left( \frac{\partial U}{\partial T} \right)_V dT + \left[ \left( \frac{\partial U}{\partial V} \right)_T + P \right] dV$$

## PART II MCQ QUESTIONS

- (4) Answer all the **TEN MCQ** questions below. There is only **ONE** answer for each question.

(40 marks)

- (a) Which of the following is **NOT** always true of a reversible adiabatic process?

- A No heat is absorbed in the process
- B No heat is released in the process
- C constant temperature in the process
- D change in internal energy,  $dU = \text{adiabatic work}$
- E it is also an isentropic process

ANS: C

- (b) Which of the following is likely to be associated with an isochoric process in an isolated system?

- A energy transferred by mechanical work
- B energy transferred by thermal interaction only
- C constant pressure condition
- D Adiabatic process
- E constant enthalpy

ANS: B

- (c) 2 objects with different mass and temperature come into contact. If the 2 objects are completely isolated from the surrounding, which of the following is likely to be true at equilibrium?

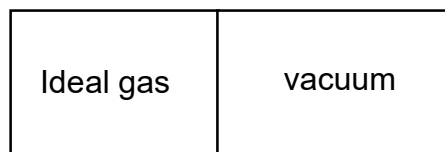
- A Their combined accessible microstate is maximum
- B Their combined temperature is maximum
- C Their respective internal energy are equal
- D The combined enthalpy is maximum
- E Work done is  $PdV$

ANS: A

- (d) A volume of gas is separated from a perfect vacuum as shown in the figure below. Given that the 2 containers are identical and insulated from the surrounding, which of the following is likely to be true if the gas is allowed to expand into the vacuum suddenly?

- A The pressure of the gas remains constant
- B The temperature of the gas will increase
- C The internal energy of the gas will be halved
- D Work done by the gas is 0
- E The volume of the gas remains constant

**ANS: D**



- (e) “in all interaction between isolated systems, the energy available to perform work will always be less than that of the initial state.” The statement is synonymous to:

- A First Law of Thermodynamics
- B Equilibrium
- C Third Law of Thermodynamics
- D Zeroth Law of Thermodynamics
- E Second Law of Thermodynamics

**ANS: E**

- (f) “the internal energy of an ideal gas is only dependent on its temperature and not the volume” The statement implies :

- A  $\left(\frac{\partial U}{\partial V}\right)_T = 0$
- B  $\left(\frac{\partial U}{\partial T}\right)_V = 0$
- C  $\left(\frac{\partial V}{\partial T}\right)_U = 0$
- D  $\left(\frac{\partial U}{\partial V}\right)_T \neq 0$
- E  $\left(\frac{\partial V}{\partial T}\right)_U \neq 0$

**ANS: A**

(g) Which of the following process may be described as the Joule-Thompson effect?

- A isentropic
- B isenthalpic
- C isothermal
- D isochoric
- E isobaric

**ANS: B**

(h) Which of the following is NOT a property of an ideal gas?

- A The internal energy is not a function of the volume
- B The enthalpy is not a function of the pressure
- C  $PV = nRT$
- D  $C_V - C_P = R$
- E No intermolecular forces

**ANS: D**

(i) At constant temperature and pressure, which of the following is NOT true?

- A the internal energy of an isolated system is constant
- B the Gibbs free energy of an isolated system is constant
- C the Helmholtz free energy of an isolated system is constant
- D the enthalpy of an isolated system is constant
- E the entropy of an isolated system is constant

**ANS: C**

(j) The rate of decrease in Helmholtz free energy with temperature at constant V equals to the \_\_\_\_\_ of the system:

A enthalpy

B internal energy

C pressure

D volume

E entropy

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| <b>ANS: E</b> |
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